

Abie Thattamparambil

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EDUCATION

Toronto Metropolitan University (Formerly Ryerson)

Toronto, ON

Department of Electrical, Computer, and Biomedical Engineering - Biomedical Engineering

Expected May 2027

- **Relevant Coursework:** Signals and Systems I & II, Biomedical Instrumentation, Circuits, Devices and Sensors, Signals and Systems, Microprocessor Systems, Control Systems and Bio-robotics, Electrical Machines and Actuators

WORK EXPERIENCE

MBG Corporate Services

Abu Dhabi, UAE

Marketing and Business Development Internship

May – August 2022

- Created a detailed **data tracking system** using **Excel** to manage information for over 100 companies. Applied **data visualization** techniques to track progress and inform management decisions.
- Initiated discussions on potential partnerships and deals through cold calls, leveraging the compiled data for targeted outreach.
- Collaborated with the business development team, creating technical presentations and supporting negotiations, emphasizing clear communication of complex technical concepts.

Biomedical Engineering Course Union (BECU)

Toronto, Canada

Corporate Associate

January 2025 – Present

- Coordinated and scheduled outreach emails to biomedical companies for lab tours, sponsorships, and career events, helping to strengthen industry ties.
- Played a key role in acquiring sponsorship deals and facilitating lab tours or info sessions that exposed students to biomedical companies.

PROJECTS

[Click here](#) to view my portfolio

Blood Alcohol Concentration Breathalyzer using MQ-3 Sensor and Arduino

January – April 2025

- Designed and built a low-cost breathalyzer using the MQ-3 alcohol sensor and Arduino to detect and calculate Blood Alcohol Content (BAC) based on BrAC values and user input.
- Calibrated sensor readings and implemented a voltage divider circuit with a Wheatstone bridge for improved measurement accuracy.

Machine Learning Model for Kidney Cancer Detection

August – September 2024

- Developed a machine learning model with 92% accuracy for identifying kidney cancer from over 10,000 CT and PET scans, applying high-speed data processing skills that are critical in AI and data networking applications..
- Employed generative AI techniques and data augmentation to enhance dataset diversity, improving the model's accuracy in distinguishing between cancerous and non-cancerous scans.

Interfacing an Analog Temperature Sensor with PIC24 Microcontroller

January – April 2023

- Designed and implemented a system to interface an analog temperature sensor (TMP26) with a PIC24 microcontroller, handling ADC, UART transmission, and real-time data monitoring—skills vital for integrated circuit layout and data transmission efficiency.
- Developed embedded firmware to monitor temperature, control a DC fan's speed, and display real-time temperature data, improving microcontroller interfacing skills. Designed and simulated the project using Proteus.

Motorized 3D-Printed Orthosis for Hand Rehabilitation

January – April 2022

- Designed and developed a low-cost, 3D-printed exoskeleton to assist patients with hand immobility due to neuromuscular diseases.
- Collaborated with a team of 4 to design and integrate EMG sensors to detect muscle contractions and control motor functions through an Arduino, showcasing sensor layout skills relevant to mixed-signal layout design and connectivity, resulting in an A+ on the project.

SKILLS

Programming: Arduino, C/C++, Python, CSS, HTML, MATLAB, Tensorflow, PyTorch

Software: MPLAB X IDE, Proteus, Multisim, Netbeans, Microsoft Office, SolidWorks